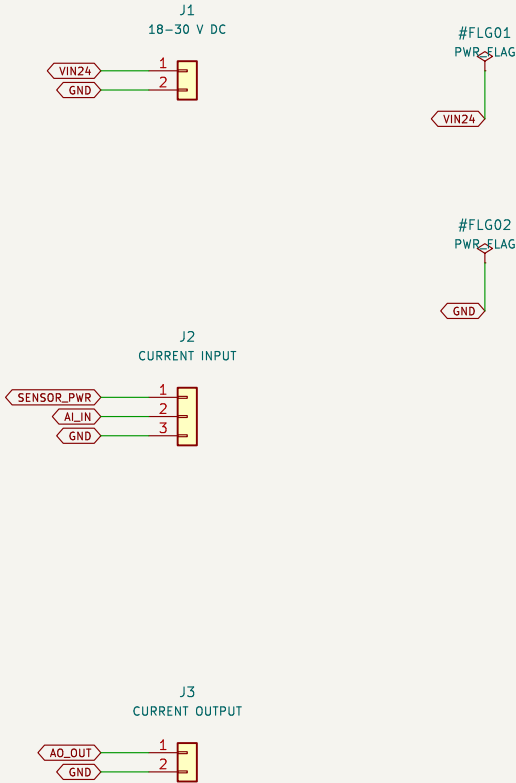


01 / SYSTEM INTERFACES

ESP32 / 24 V / 4–20 mA | Rev A – engineering prototype | 2026–09–07

FIELD TERMINALS

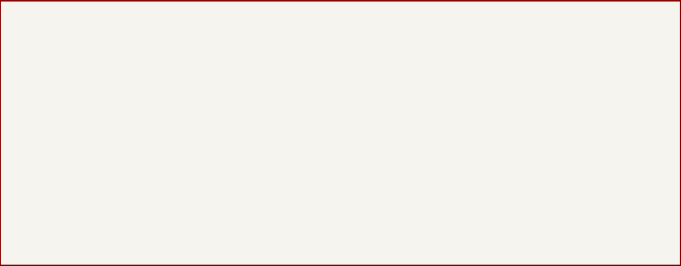
All returns and USB ground are common. No galvanic isolation.



INPUT: two-wire sensor + to J2.1, sensor – to J2.2.
Externally powered source: signal to J2.2, return to J2.3.
OUTPUT: passive receiver + to J3.1, receiver – to J3.2.
Do not connect AO to an actively powered loop input.

VALIDATION TARGETS
Room-temperature accuracy after calibration: +/-16 uA, 10 samples/s.
Bench-check miswire response, startup transients and loop compliance.
No IEC surge qualification or temperature accuracy claim at this revision.

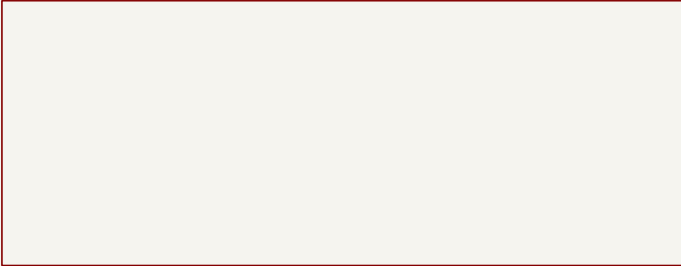
02 / POWER AND SENSOR SUPPLY



File: power.kicad_sch

VIN24 -> protected 24 V + 3.3 V
SENSOR_PWR: approximately 40 mA limit

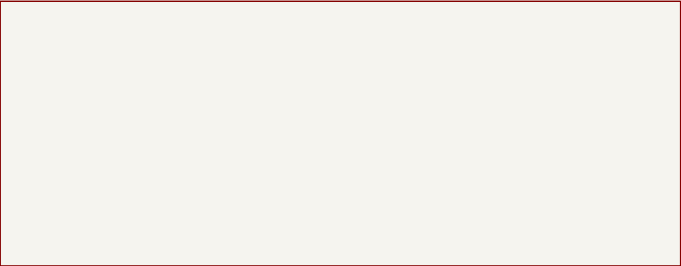
03 / ESP32 AND USB



File: controller.kicad_sch

USB-C programming: 24 V board power required
I2C, output enable and fault monitoring

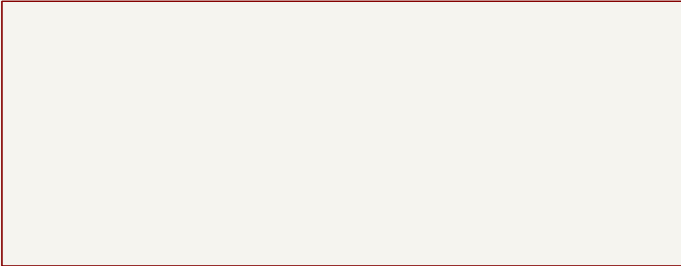
04 / PROTECTED CURRENT INPUT



File: analog_input.kicad_sch

AI_IN -> limiter -> 100 ohm shunt -> ADC
Nominal range: 4–20 mA; measure 0–25 mA

05 / PRECISION CURRENT OUTPUT



File: analog_output.kicad_sch

DAC -> XTR111 -> protected sourcing output
4–20 mA into 0–500 ohms, including cable

Engineering review required before fabrication
ESP32 Loop I/O

Sheet: /
File: chatgpt_astra_test2.kicad_sch

Title: 01 / SYSTEM INTERFACES

Size: A3 | Date: 2026–09–07
KiCad E.D.A. 9.0.7

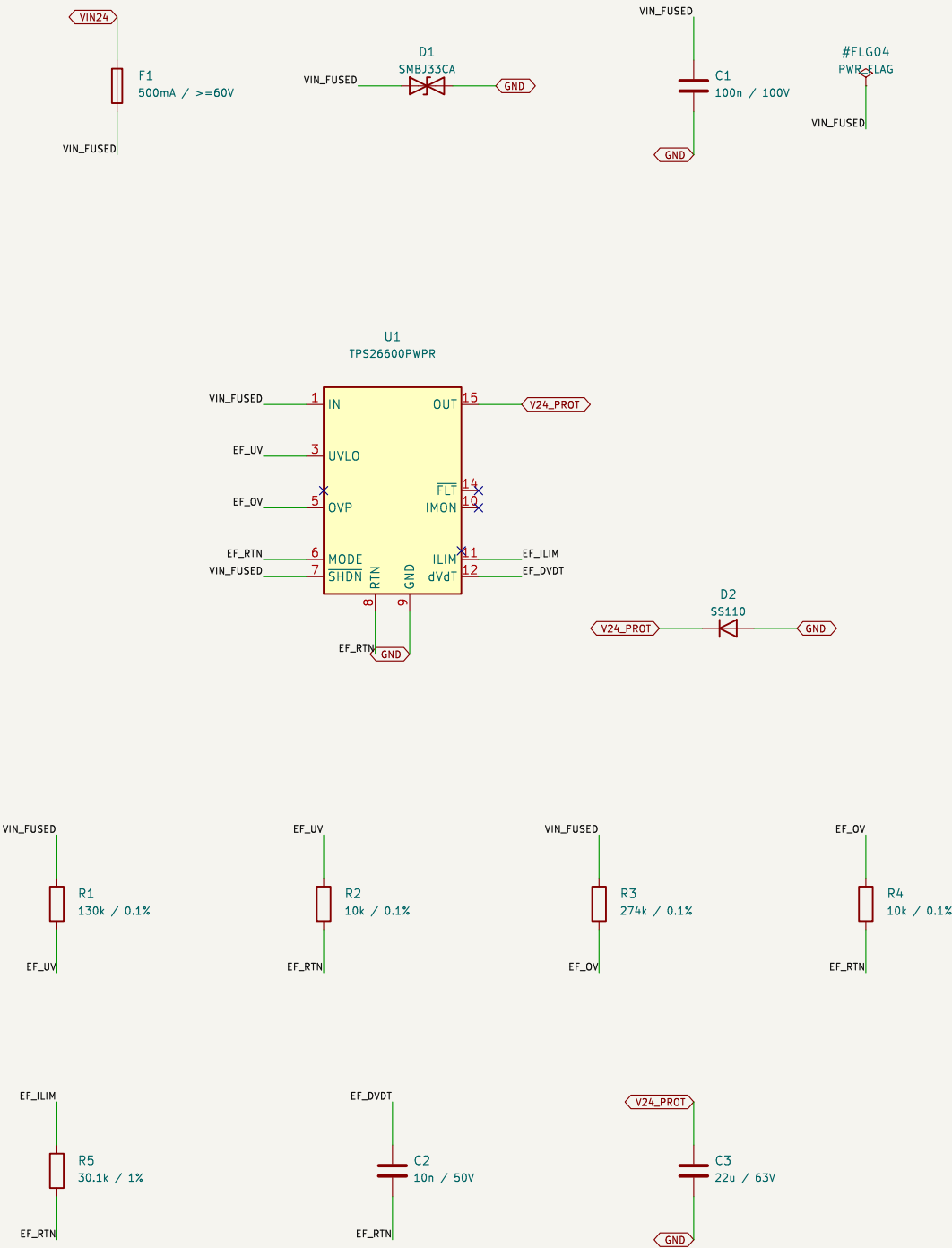
Rev: A – prototype
Id: 1/5

02 / POWER AND SENSOR SUPPLY

ESP32 / 24 V / 4–20 mA | Rev A – engineering prototype | 2026–09–07

24 V ENTRY / eFUSE

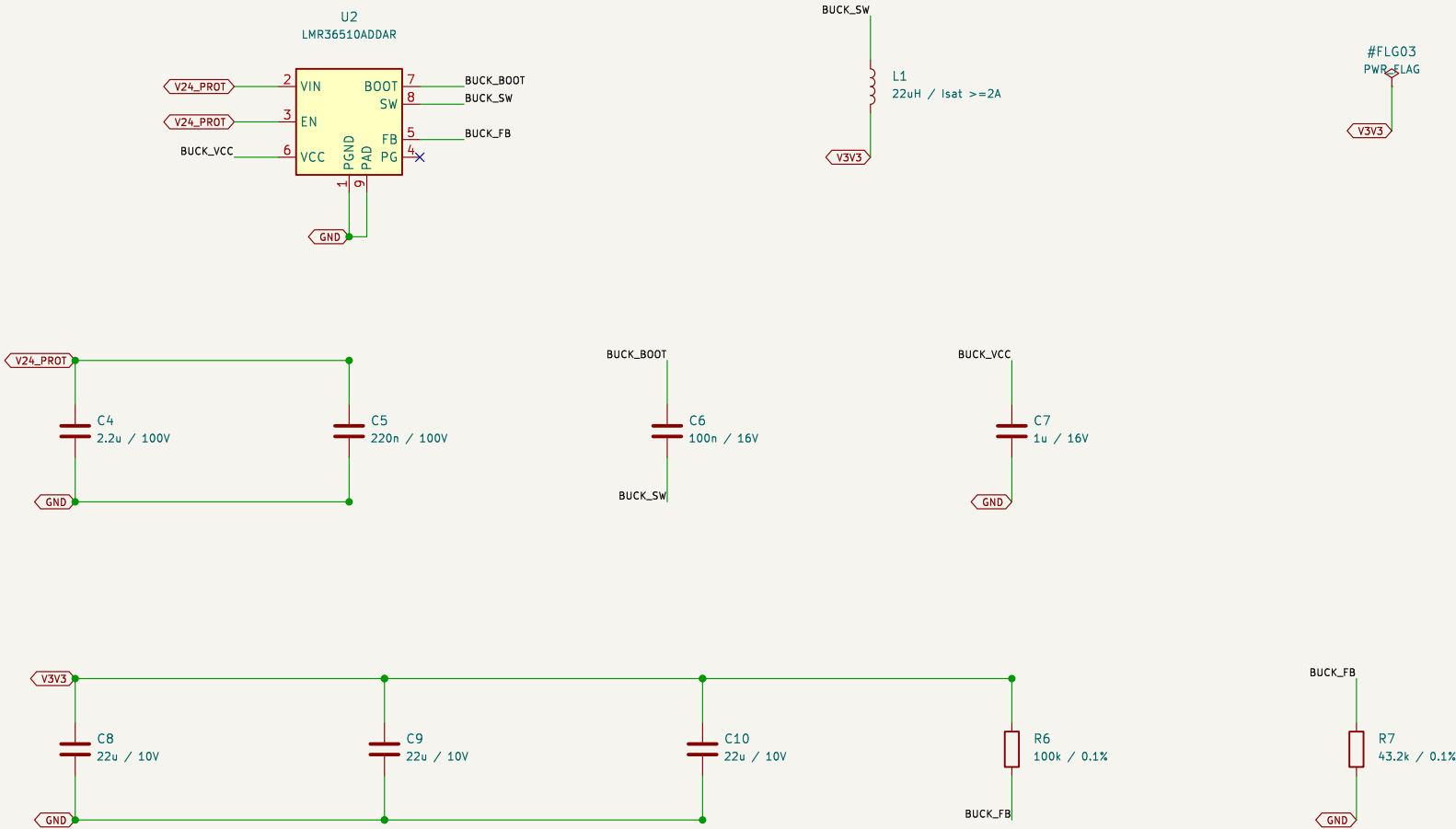
RTN is the internal reference: never short it to system GND.



UV rising approx. 16.66 V; OV rising approx. 33.80 V.
Current limit approx. 399 mA. Auto-retry current limiting.
FLT unused: avoids reverse-input coupling into MCU ground.

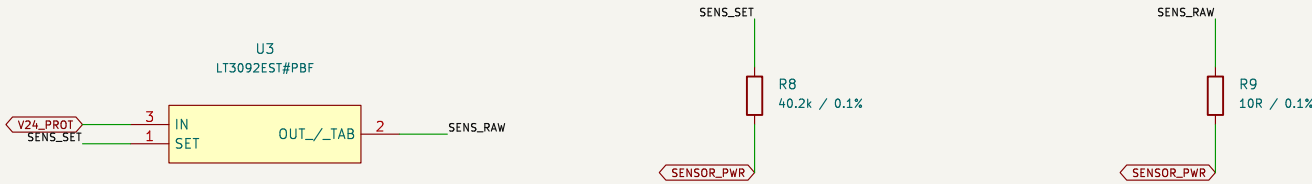
3.3 V / 1 A BUCK

400 kHz. Short VIN–CIN–PGND loop; minimize switch–node copper.



SENSOR SUPPLY / 40 mA LIMIT

All LT3092 current returns through its load; no ground branch.



Engineering review required before fabrication

ESP32 Loop I/O

Sheet: /02 / POWER AND SENSOR SUPPLY/

File: power.kicad_sch

Title: 02 / POWER AND SENSOR SUPPLY

Size: A3

Date: 2026–09–07

Rev: A – prototype

KiCad E.D.A. 9.0.7

Id: 2/5

03 / ESP32 AND USB

ESP32 / 24 V / 4–20 mA | Rev A – engineering prototype | 2026–09–07

ESP32–S3–WROOM–1–N8

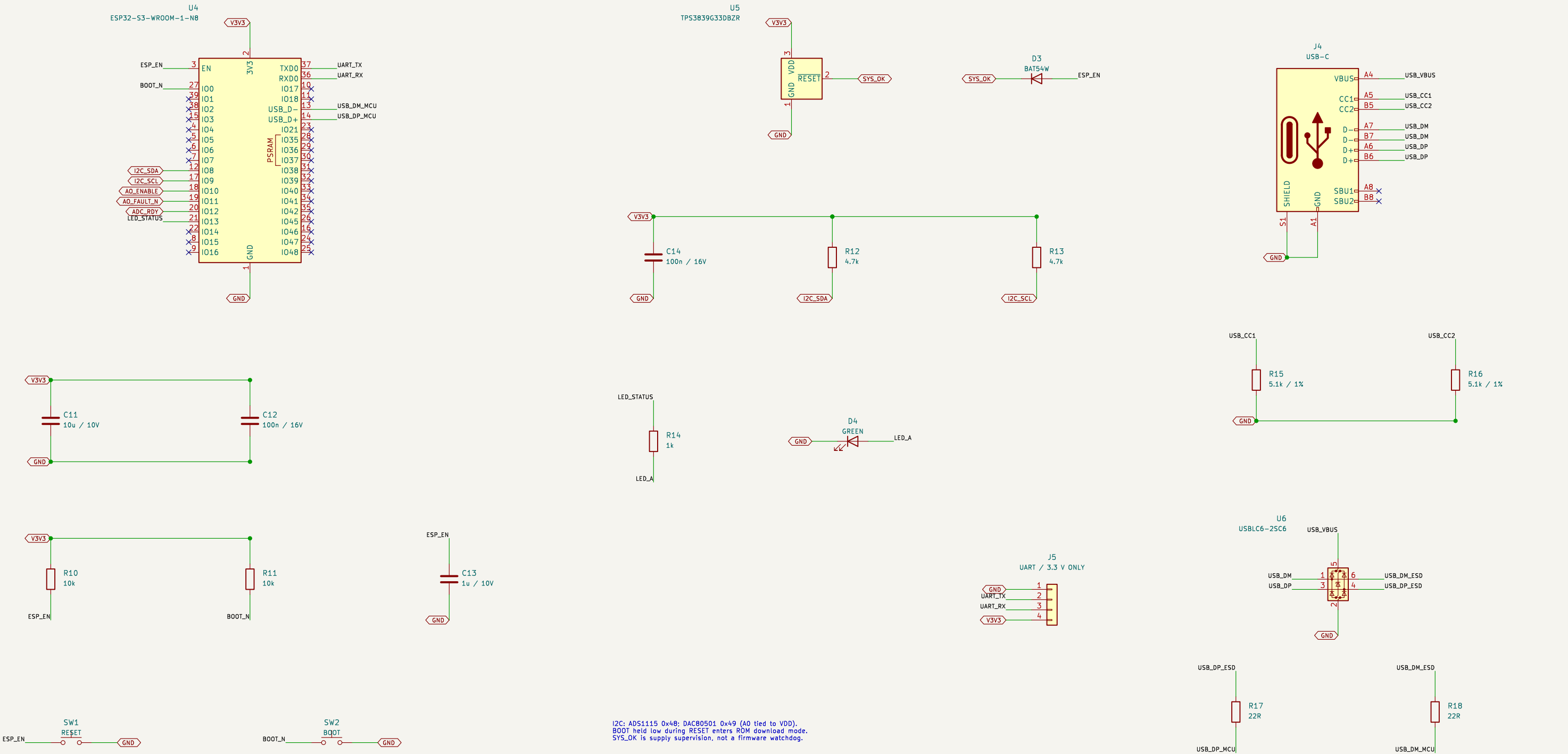
No PSRAM. GPIO19/20 reserved for native USB. Antenna keepout required.

SUPERVISION / I2C / DEBUG

SYS_OK disconnects ADC and inhibits A0 below 3.08 V nominal.

USB–C / DATA ONLY

No VBUS–to–board power connection.

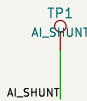
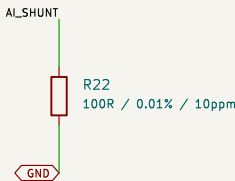
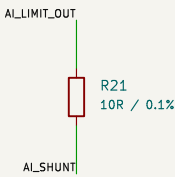
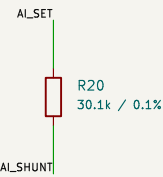
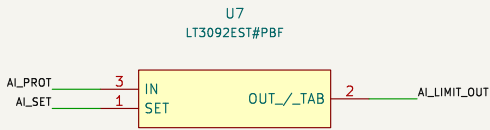
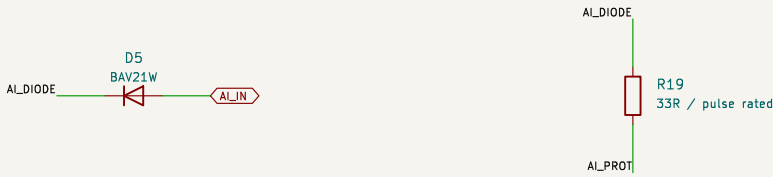


04 / PROTECTED CURRENT INPUT

ESP32 / 24 V / 4–20 mA | Rev A – engineering prototype | 2026–09–07

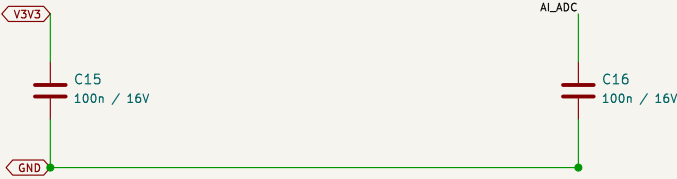
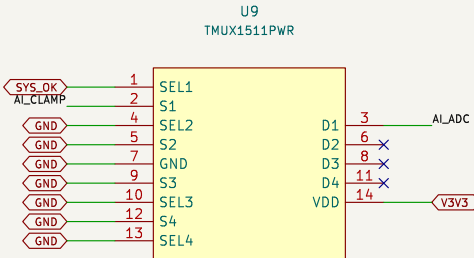
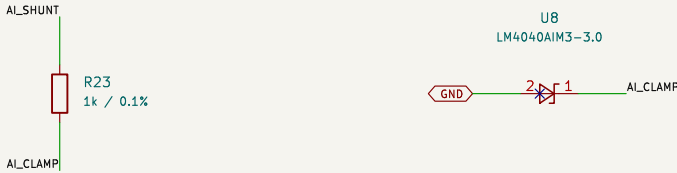
FIELD INPUT / 30 mA LIMIT

Two-terminal limiter preserves DC loop current through shunt.



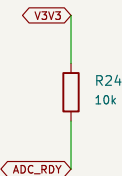
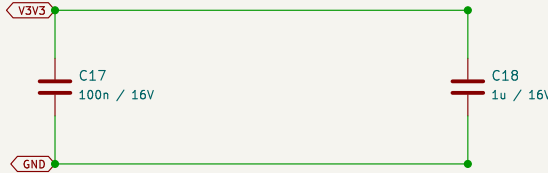
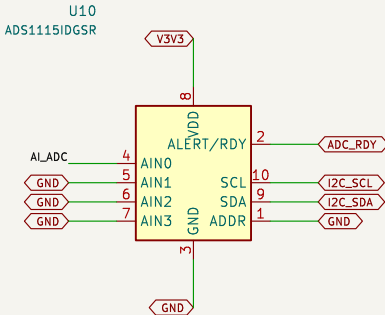
ADC FAULT CLAMP / DISCONNECT

3 V shunt clamp is independent of the board supply.



16-BIT ADC / I2C 0x48

PGA +/-4.096 V; AIN0 single-ended. Average for 10 Hz reporting.



100 ohm; 0.4 V @ 4 mA; 2 V @ 20 mA.
Nominal ADC step: 1.25 uA at +/-4.096 V.
Calibration includes ADC loading and clamp leakage.
Check turn-on surge and clamp recovery on bench.

Engineering review required before fabrication

ESP32 Loop I/O

Sheet: /04 / PROTECTED CURRENT INPUT/

File: analog_input.kicad_sch

Title: 04 / PROTECTED CURRENT INPUT

Size: A3

Date: 2026–09–07

Rev: A – prototype

KiCad E.D.A. 9.0.7

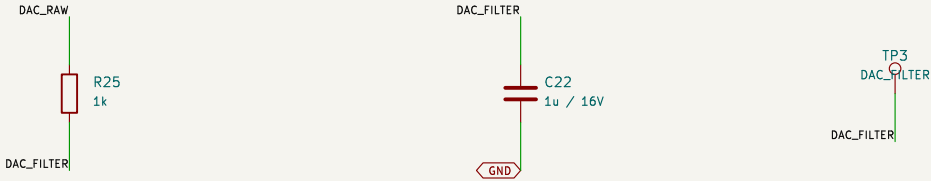
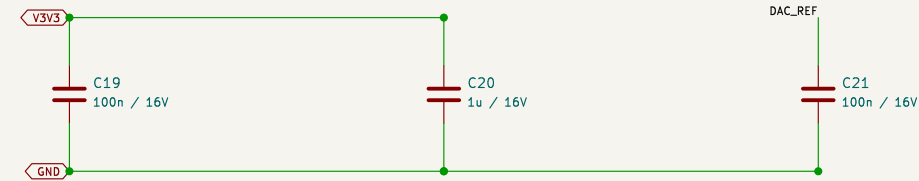
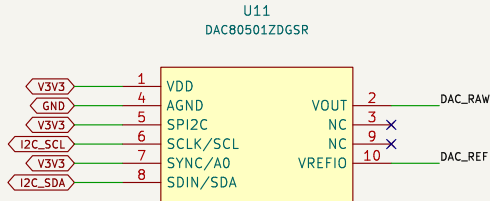
Id: 4/5

05 / PRECISION CURRENT OUTPUT

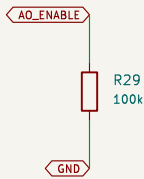
ESP32 / 24 V / 4–20 mA | Rev A – engineering prototype | 2026–09–07

16-BIT DAC / I2C 0x49

Zero-scale startup. REF-DIV=1, BUFF-GAIN=1 -> 2.5 V range.

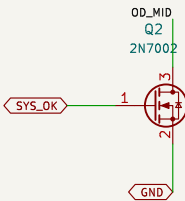
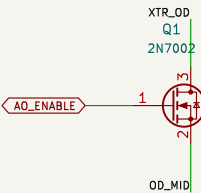
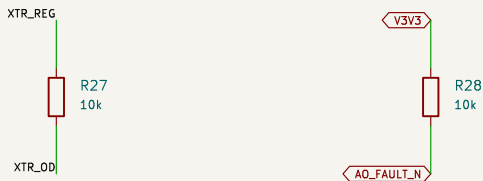
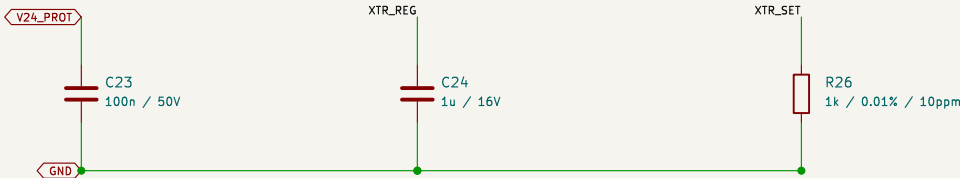
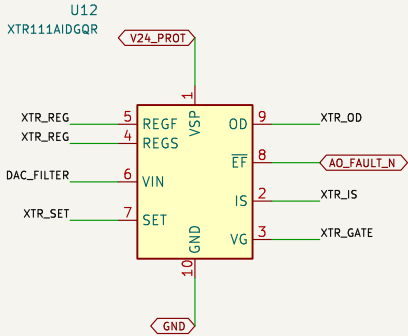


$I_{out} = 10 \cdot V_{IN} / R_{SET}$. $R_{SET} = 1.00 \text{ kohm}$.
0.4 V -> 4 mA; 2.0 V -> 20 mA.
Firmware limits normal commands to calibrated 4–20 mA.
DAC range allows diagnostic current up to 25 mA.



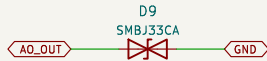
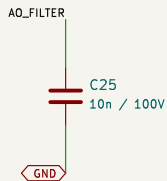
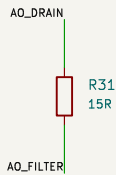
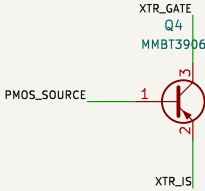
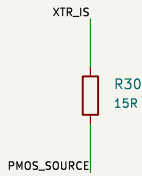
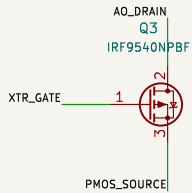
XTR111 / DEFAULT DISABLED

OD pull-up uses XTR local regulator, independent of MCU power.



OUTPUT PASS STAGE / TERMINAL

100 V P-MOSFET; T0–220 provides prototype thermal margin.



PNP limit approx. 33–37 mA; not a precision setpoint.
D8 blocks external positive loop power.
Negative AO miswire increases MOSFET dissipation.
Verify SDA, startup glitch and capacitive-load stability.